An Improved Upper Bound for Three-Robot Wireless Evacuation on the Disk

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Abstract

We study the wireless three-robot evacuation problem on the unit disk and prove an upper bound of $1 + \frac{2\pi}{3} - \frac{y_{\text{opt}}}{2} + \sqrt{3} \approx 4.2185170$, where y_{opt} is the unique root in $(0, \pi/6)$ of a transcendental master equation derived below. This improves the best known upper bound of Czyzowicz et al. [2] (≈ 4.2192973) by approximately $0.00078 \approx 1/1280$, and leaves a gap of about 0.06 against the best known lower bound of the same paper. The improvement is obtained by retaining the structural skeleton of the A_3 algorithm of [2] and rebalancing a single deployment parameter y so that *four* symmetric worst-case exit angles bind simultaneously, rather than the two balanced by Czyzowicz et al. Rigour is provided by an analytic zone-by-zone enumeration of the evacuation time's worst-case regimes; a 40-digit integer-relation search (PSLQ) over the usual atomic constants rules out elementary closed-form reductions of y_{opt} below a specified norm radius.

1 Introduction

Robot evacuation from the disk is the following problem. A team of mobile robots, each moving at unit speed, starts at the centre of the unit disk. An exit sits at an unknown point on the boundary circle. The goal is a set of trajectories — one per robot, chosen in advance — that minimises the worst-case time (over the placement of the exit) until every robot reaches the exit. A robot "finds" the exit the first moment its trajectory visits that boundary point; under the *wireless* communication model, we assume the finder broadcasts instantaneously and every robot then walks the chord from wherever it currently is to the exit.[1]

Figure 1 illustrates the setup. Czyzowicz et al. [2] introduced the problem for general k and proved an asymptotic upper bound $3 + \pi/k + O(k^{-4/3})$ valid for $k \geq 100$. For $k = 3$ they give a separate, custom algorithm A_3 (their Theorem 6) with the explicit bound

$$T_{\text{CGK}} \;:=\; \tfrac{4\pi}{9} + \tfrac{2\sqrt{3}+5}{3} + \tfrac{1}{600} \;\approx\; 4.21929727, \tag{1}$$

and a matching lower bound of approximately 4.15935 (their Theorem 7). The gap of ≈ 0.06 has been open since 2014.

Contribution. Our main result (Theorem 7 below) proves the improved upper bound

$$T(y_{\text{opt}}) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y_{\text{opt}}}{2} + \sqrt{3} \;\approx\; 4.21851699\ldots, \tag{2}$$

[1] Other communication models in the literature include *face-to-face* (F2F), in which robots learn of a discovery only by co-location [1]; and asymmetric models such as *sender/receiver* [4] and *priority evacuation* [3]. Here we treat only the wireless case.

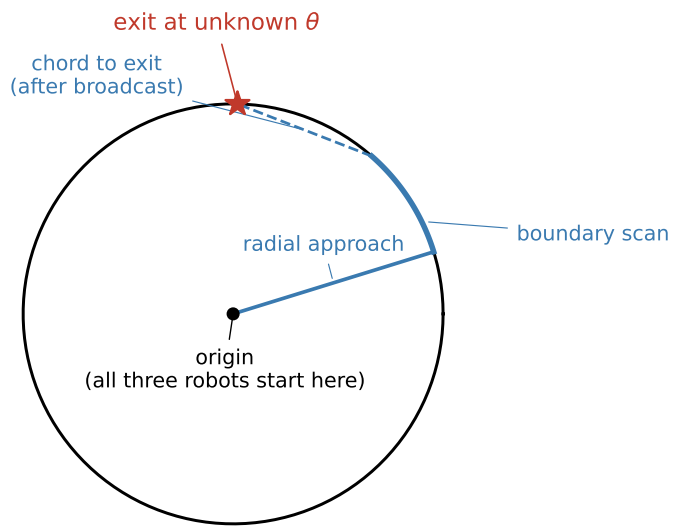

Figure 1: The $k = 3$ wireless evacuation problem. Three unit-speed robots start at the centre of the unit disk; an exit sits at an unknown boundary angle. A robot's trajectory typically has a radial approach phase to the boundary, a boundary-scan phase, and — after an exit is broadcast — a straight chord to the exit.

for a specific deployment parameter y_{opt} determined by a transcendental equation (Theorem 8 and remark 9). The optimum is reached when four symmetric worst-case exit angles simultaneously bind. The result is proved by an analytic zone decomposition of the evacuation time (Lemmas 5 and 6); no lower-bound improvement is claimed.

High-level strategy. We retain Czyzowicz et al.'s A_3 structural skeleton verbatim, but treat the single angular offset y between the two deployment points on the boundary as a tunable parameter. In [2] this is set to $y_{\text{paper}} = \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{401}{300}$ by a hand-tuned argument balancing two worst-case regimes. A third worst-case regime exists (Proposition 4), and balancing all three yields a slightly smaller y and the improved bound. The $1/600$ slack in (1) is in fact exactly half the penalty imposed by ignoring this third regime; our optimum removes it and improves by a further $\approx 1/(1280)$ beyond.

Provenance. This paper is the principal positive result of *evacuation-claude*, a project using Anthropic's Claude as a sustained research collaborator to survey open gaps across four reference papers on robot evacuation. Other probes in the same project — an ε-cut refinement of Brandt et al.'s 2-robot F2F algorithm [1], and a bracket for priority evacuation with a single servant [3] — yielded a null result and a loose bracket respectively, leaving the $k = 3$ wireless upper bound as the gap the tooling could tighten. Source, experimental scripts, and per-session research log are at `https://github.com/jaredraycoleman/evacuation-claude`.

2 Preliminaries

The closed unit disk and its boundary. Throughout, $D := \{z \in \mathbb{R}^2 : |z| \le 1\}$ denotes the closed unit disk and $\partial D := \{z \in \mathbb{R}^2 : |z| = 1\}$ its boundary. We identify a boundary point with its angle $\theta \in [0, 2\pi)$ via $e(\theta) := (\cos\theta, \sin\theta)$. The *shorter-arc distance* between $e(\alpha)$ and $e(\beta)$ on ∂D is $\min(|\alpha - \beta|,\ 2\pi - |\alpha - \beta|)$; the corresponding *chord length* is $|e(\alpha) - e(\beta)| = 2|\sin\big((\alpha - \beta)/2\big)|$.

Robots, trajectories and algorithms. A *trajectory* is a 1-Lipschitz map $\rho : [0, \infty) \to D$ with $\rho(0) = 0$; the Lipschitz constant 1 encodes unit-speed motion. An *algorithm* for the k-robot evacuation problem is a tuple $\mathcal{A} = (\rho_1, \ldots, \rho_k)$ of trajectories (the robots' pre-committed behaviours in the absence of any exit report), together with the reactive rule below. We require that $\mathcal{A}$ *covers the boundary*, meaning

$$\forall \theta \in [0, 2\pi) : \quad \min_i \inf\{t \ge 0 : \rho_i(t) = e(\theta)\} < \infty.$$

Wireless communication and evacuation time. Fix an exit $e(\theta) \in \partial D$. For algorithm $\mathcal{A}$:

(a) The *discovery time* of exit θ is

$$t_{\text{find}}(\theta) \ := \ \min_i \inf\{t : \rho_i(t) = e(\theta)\}.$$

A robot attaining this infimum is called a *finder* (a choice in case of ties).

(b) At time $t_{\text{find}}(\theta)$ the finder broadcasts instantaneously and every robot switches to the straight-line chord at unit speed from its current position to $e(\theta)$. Robot i therefore reaches the exit at time

$$T_i(\theta) \ := \ t_{\text{find}}(\theta) + \big|\rho_i(t_{\text{find}}(\theta)) - e(\theta)\big|.$$

(c) The *evacuation time* for exit θ is

$$\mathrm{evac}_{\mathcal{A}}(\theta) \;:=\; \max_i T_i(\theta) \;=\; t_{\mathrm{find}}(\theta) + \max_i \big|\rho_i(t_{\mathrm{find}}(\theta)) - e(\theta)\big|. \tag{3}$$

A robot attaining this maximum is called the *bottleneck* for θ.

The *worst-case evacuation time of* $\mathcal{A}$ is $T(\mathcal{A}) := \sup_{\theta\in[0,2\pi)} \mathrm{evac}_{\mathcal{A}}(\theta)$. The *$k$-robot wireless evacuation problem on the disk* asks for $T_k^* := \inf_{\mathcal{A}} T(\mathcal{A})$ over k-robot algorithms. This paper concerns $k = 3$.

Scanning. For us a *boundary-scan phase* is a subinterval $[t_1, t_2] \subseteq [0, \infty)$ on which a trajectory ρ satisfies $\rho(t) \in \partial D$ and $t \mapsto \rho(t)$ traverses a contiguous boundary arc at unit speed. The arc covered is automatically of arc-length $t_2 - t_1$. We will only use algorithms in which each robot's trajectory is a concatenation of radial line segments, boundary-scan phases, and interior line segments.

3 The $A_3(y)$ algorithm family

The Czyzowicz et al. algorithm A_3 of [2] depends on a deployment parameter $y \in (0, 2\pi/3)$. We treat y as a free parameter and write $A_3(y)$ for the corresponding algorithm; the specific value $y = \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{401}{300}$ recovers the algorithm analysed in [2].

Definition 1 ($A_3(y)$)**.** Fix $y \in (0, 2\pi/3)$ and set $L := \pi - y/2$. Place

$$A := e(0), \quad B := e(2\pi - y), \quad C := e(\pi - y/2).$$

The three robots' trajectories are:

ρ_1: radial $0 \to A$ on $t \in [0, 1]$; boundary scan CCW from A on $t \in [1, 1 + L]$; parked at $e(L)$ for $t \geq 1 + L$.

ρ_2: radial $0 \to B$ on $[0, 1]$; boundary scan CW from B on $[1, 1+L]$; parked at $e(2\pi - y - L) = e(L)$ for $t \geq 1 + L$.

ρ_3: radial $0 \to B$ on $[0, 1]$; boundary scan CCW from B on $[1, 1+y]$, ending at $e(2\pi) = A$; straight line from A to the origin on $[1 + y,\, 2 + y]$; radial $0 \to C$ on $[2 + y,\, 3 + y]$; parked at C for $t \geq 3 + y$.

The reactive rule (chord to the exit upon broadcast) is the standard one from Section 2.

The three scan phases traverse arcs $[0, L]$, $[L, 2\pi - y]$, and $[2\pi - y, 2\pi)$ of lengths L, L, and y respectively. Since $2L + y = 2\pi$, these arcs partition $[0, 2\pi)$ and $A_3(y)$ covers the boundary. In particular every exit angle $\theta \in [0, 2\pi)$ is found by exactly one robot during its scan phase. Figure 2 shows the resulting trajectories at the value $y = y_{\mathrm{opt}}$ that will emerge below.

We write $T(y) := T(A_3(y))$ for the worst-case evacuation time.

Let $T(y)$ denote the worst-case evacuation time of A_3 parametrised by y. Both [2] and the present note study $T(y)$ for y near $y_\star := \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{4}{3}$.

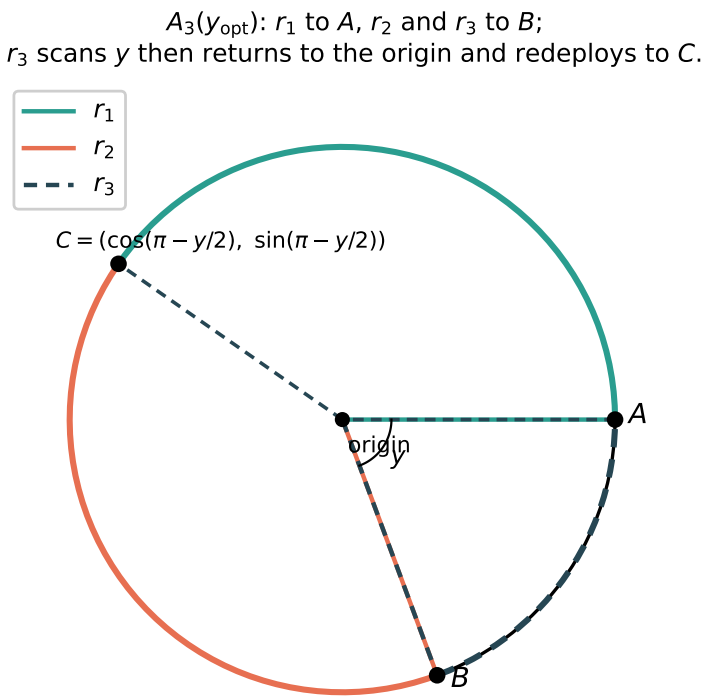

Figure 2: The A_3 algorithm at $y = y_{\text{opt}}$. Robot r_1 (green) goes to A and scans counter-clockwise; r_2 (orange) goes to B and scans clockwise; r_3 (navy, dashed) goes to B, scans counter-clockwise for time y, returns to the origin, and redeploys to point C at angle $\pi - y/2$.

4 Main result

Throughout this section $\mathrm{evac}(\theta) := \mathrm{evac}_{A_3(y)}(\theta)$ with dependence on y implicit. Our main theorem exhibits a specific deployment parameter y_{opt} and proves

$$T(y_{\mathrm{opt}}) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y_{\mathrm{opt}}}{2} + \sqrt{3} \;\approx\; 4.21851699\ldots.$$

The scalar y_{opt} is characterised explicitly by a transcendental master equation (Theorem 8 below). The proof has three ingredients: a case analysis identifying three families of candidate worst-case exit angles (Propositions 2 to 4); a zone decomposition of $[0, 2\pi)$ on which evac admits a closed form (Lemma 5); and a zone-wise maximisation (Lemma 6) whose outcome identifies $T_1 = T_3$ as the global maximum at $y = y_{\mathrm{opt}}$.

4.1 Three candidate worst cases

Set $L := \pi - y/2$; assume $y \in (0, 2\pi/3)$ so $L \in (2\pi/3, \pi)$.

Proposition 2 (Case 1: ρ_2 bottleneck)**.** *At $\theta_1(y) := 2\pi/3 - y/2 \in (0, L)$ the exit is found by ρ_1 during its scan phase and the bottleneck is ρ_2 at chord $\sqrt{3}$, giving*

$$T_1(y) \;:=\; \mathrm{evac}(\theta_1(y)) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y}{2} + \sqrt{3}. \tag{4}$$

Proof. Since $\theta_1 \in (0, L]$, the exit lies in ρ_1's scan arc, so $t_{\mathrm{find}}(\theta_1) = 1+\theta_1$ and ρ_1 is at $e(\theta_1)$ (chord 0). At time $1+\theta_1$, ρ_2 has been scanning clockwise from B for θ_1 time units and is at angle $(2\pi - y) - \theta_1$. Its chord to the exit has length

$$2\left|\sin\tfrac{(2\pi-y-\theta_1)-\theta_1}{2}\right| \;=\; 2\sin\!\left(\pi - \tfrac{y}{2} - \theta_1\right) \;=\; 2\sin\!\left(\tfrac{y}{2} + \theta_1\right),$$

using $\sin(\pi - x) = \sin x$ (valid since $y/2 + \theta_1 \in (0, \pi)$). Plugging $\theta_1 = 2\pi/3 - y/2$ gives $2\sin(2\pi/3) = \sqrt{3}$. We verify in Lemma 6(2) that ρ_3's chord at this θ is strictly less than $\sqrt{3}$ for y near y_{opt}, so the bottleneck is ρ_2 and $\mathrm{evac}(\theta_1) = 1 + \theta_1 + \sqrt{3}$, which simplifies to (4). □

Proposition 3 (Case 2: ρ_3 bottleneck at the arc midpoint)**.** *At $\theta_2(y) := L$ (the common endpoint of ρ_1's and ρ_2's scan arcs) the exit is reached simultaneously by ρ_1 and ρ_2 at time $1 + L$; ρ_3 is at fraction $\pi - 3y/2 - 1$ along its redeploy chord to C, and*

$$T_2(y) \;:=\; \mathrm{evac}(\theta_2(y)) \;=\; (1 + L) + \big(1 - (\pi - 3y/2 - 1)\big) \;=\; 3 + y. \tag{5}$$

Proof. Since L is the CCW-end of ρ_1's scan and the CW-end of ρ_2's, $t_{\mathrm{find}}(L) = 1 + L$ and both are at $e(L)$ with chord 0. For ρ_3 at time $1 + L$, we check $1 + L \in (2 + y,\, 3 + y)$, i.e., $\pi - y/2 \in (1 + y,\, 2 + y)$, i.e., $y \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right) \approx (0.76,\, 1.40)$, which contains $y_{\mathrm{opt}} \approx 1.216$. Hence ρ_3 is in its origin-to-C redeploy phase: $\rho_3(1 + L) = (1 + L - (2 + y)) \cdot C = (\pi - 3y/2 - 1) \cdot C$. Its chord to the exit C is $1 - (\pi - 3y/2 - 1) = 2 - \pi + 3y/2$. Summing discovery time and chord gives $T_2(y) = 1 + L + (2 - \pi + 3y/2) = 3 + y$. □

A third regime involves an implicit interior maximum in θ.

Proposition 4 (Case 3: ρ_3 bottleneck mid-redeploy)**.** *For $y \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right)$ (in particular at y_{opt}), set*

$$T_3(y, \theta) \;:=\; 1 + \theta + \big|\rho_3(1 + \theta) - e(\theta)\big|, \qquad \theta \in [1 + y,\, L).$$

Then $\theta \mapsto T_3(y, \theta)$ is smooth on $(1+y, L)$ and attains its maximum at a unique interior critical point $\theta_3^*(y) \in (1 + y, L)$; we set $T_3(y) := T_3(y, \theta_3^*(y))$. The explicit form of $\theta_3^*(y)$ is given in Theorem 8.

Proof. On $[1+y,\, L]$ the discovery time $1+\theta$ lies in $[2+y,\, 1+L] \subseteq [2+y,\, 3+y]$ (using $L < 2+y$, which holds for y in the stated range). Hence ρ_3 is in its redeploy phase and $\rho_3(1+\theta) = (\theta - 1 - y) \cdot e(L)$ is linear in θ. The chord $|\rho_3(1+\theta) - e(\theta)|$ is therefore a smooth positive function of θ, and so is $T_3(y, \theta)$. Existence and uniqueness of the critical point follow from Theorem 8 below. □

T_1 is strictly decreasing in y, T_2 strictly increasing, so they cross at $y_\star := \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{4}{3} \approx 1.2176$. The value $y_{\text{paper}} = y_\star - \frac{1}{300}$ used in [2] sits slightly below $y_\star$; in effect their analysis bounds $T_3(y_{\text{paper}})$ by $T_1(y_\star) + \frac{1}{600}$, giving the published bound $T_1(y_\star) + \frac{1}{600} \approx 4.21930$. Our Theorem 7 picks y_{opt} so that $T_1 = T_3$ *exactly* (with T_2 slack), a strictly tighter balance.

4.2 Zone decomposition

Lemma 5 (Zone decomposition)**.** *For $y \in \left(\frac{2(\pi-2)}{3},\, \frac{4\pi}{9}\right)$, the interval $[0, 2\pi)$ splits into three* finder zones*:*

$$\underbrace{[0,\, L]}_{\text{Zone 1: } \rho_1 \text{ finds}} \quad \cup \quad \underbrace{[L,\, 2\pi - y]}_{\text{Zone 2: } \rho_2 \text{ finds}} \quad \cup \quad \underbrace{[2\pi - y,\, 2\pi)}_{\text{Zone 3: } \rho_3 \text{ finds}} .$$

Discovery times are $t_{\text{find}}(\theta) = 1 + \theta$ on Zone 1, $1 + (2\pi - y - \theta)$ on Zone 2, and $1 + (\theta - 2\pi + y)$ on Zone 3. Zone 1 further splits by the phase of ρ_3 at time $1 + \theta$:

- ***1a*** *($\theta \in [0,\, y]$): ρ_3 is scanning on the boundary at angle $(2\pi - y) + \theta \pmod{2\pi}$.*
- ***1b*** *($\theta \in [y,\, 1 + y]$): ρ_3 is on its boundary-to-origin transit, at position $(1 + y - \theta,\, 0)$.*
- ***1c*** *($\theta \in [1 + y,\, L]$): ρ_3 is on its origin-to-C redeploy, at position $(\theta - 1 - y) \cdot e(L)$.*

The problem is invariant under the reflection $\theta \mapsto (2\pi - y) - \theta$ exchanging $\rho_1 \leftrightarrow \rho_2$; Zone 2 is the image of Zone 1 under this reflection.

Proof. The finder-zone partition follows from Definition 1: ρ_1's scan arc is $[0, L]$, ρ_2's is $[L, 2\pi - y]$, ρ_3's is $[2\pi - y, 2\pi)$, and these tile $[0, 2\pi)$ because $2L + y = 2\pi$. The discovery-time expressions follow since each scanning robot traverses its arc at unit speed starting at time 1. For the sub-zones: ρ_3 has phases $[0, 1]$ (approach), $[1, 1 + y]$ (scan), $[1 + y,\, 2 + y]$ (boundary-to-origin), $[2 + y,\, 3 + y]$ (origin-to-C), $[3 + y, \infty)$ (parked). So for $\theta \in [0, L]$ with $1 + \theta \in [1,\, 1 + L]$, we have $1 + \theta \in [1, 1 + y]$ iff $\theta \in [0, y]$ (sub-zone 1a); $1 + \theta \in [1 + y, 2 + y]$ iff $\theta \in [y, 1 + y]$ (1b); $1 + \theta \in [2 + y, 3 + y]$ iff $\theta \in [1 + y, 2 + y]$, which intersected with $[0, L]$ gives $[1 + y, L]$ (1c), using $L \leq 2 + y$ (equivalent to $y \geq 2(\pi - 2)/3$). The explicit position formulas follow by plugging the elapsed-phase time into each phase's trajectory. Reflection invariance is immediate from the symmetry of Definition 1. □

4.3 Zone-wise maxima at y_{opt}

For compactness write $c_i(\theta) := |\rho_i(t_{\text{find}}(\theta)) - e(\theta)|$ for $i \in \{1, 2, 3\}$, so that $\text{evac}(\theta) = t_{\text{find}}(\theta) + \max_i c_i(\theta)$ by (3).

Lemma 6 (Zone maxima at y_{opt})**.** *Let $y_{\text{opt}} \in \left(\frac{2(\pi-2)}{3},\, \frac{4\pi}{9}\right)$ be the parameter defined in Theorem 8, and set $T_1 := T_1(y_{\text{opt}})$, $T_3 := T_3(y_{\text{opt}})$.*

1. Zone 1a. $\text{evac}(\theta) = 1 + \theta + 2\sin(y/2 + \theta)$ *on $[0, y]$ and is strictly increasing. Hence* $\max_{\theta \in [0,y]} \text{evac}(\theta) = 1 + y + 2\sin(3y/2) \approx 4.1522 < T_1$ *at $y = y_{\text{opt}}$.*

2. Zone 1b. *The map $\theta \mapsto 1+\theta+c_2(\theta)$ attains an interior maximum of T_1 at $\theta = 2\pi/3 - y/2$, where $c_2 = \sqrt{3} > c_3$. On the c_3-dominant sub-interval, $1+\theta+c_3(\theta)$ has no interior critical point, and its maximum at the endpoint $\theta = 1+y$ equals $3+y_{\text{opt}} < T_1$. Consequently* $\max_{\theta\in[y,\,1+y]} \text{evac}(\theta) = T_1$.

3. Zone 1c. *$c_3(\theta) > c_2(\theta)$ throughout $[1+y,\, L]$. The function $\theta \mapsto 1+\theta+c_3(\theta)$ has a unique interior critical point at $\theta_3^*(y_{\text{opt}})$, with value $T_3 = T_1$. Both endpoints give $3+y_{\text{opt}} < T_1$.*

4. Zone 3. *$c_1 = c_2 = 2\sin(y/2)$ is constant and $c_3 = 0$; the maximum is $1+y+2\sin y \approx 4.0912 < T_1$.*

Proof. **Zone 1a.** The finder ρ_1 is at the exit, so $c_1 = 0$. By Lemma 5(1a), ρ_3 is on the boundary at angle $(2\pi - y)+\theta$, so $c_3 = 2|\sin((2\pi-y)/2)| = 2\sin(y/2)$ (independent of θ). ρ_2 is still scanning at angle $(2\pi - y) - \theta$, so as computed in the proof of Proposition 2, $c_2 = 2\sin(y/2+\theta)$. For $\theta \in (0, y]$ and $y < \pi/2$, $\sin(y/2+\theta) > \sin(y/2)$, so the bottleneck is ρ_2 and $\text{evac}(\theta) = 1+\theta+2\sin(y/2+\theta)$. Its derivative is $1+2\cos(y/2+\theta)$; on $[0, y]$ the argument $y/2+\theta$ ranges in $[y/2, 3y/2] \subset (0, \pi)$, and at $y = y_{\text{opt}} \approx 1.216$ we have $\cos(3y/2) \approx -0.25 > -1/2$, so the derivative is positive throughout. Hence evac is strictly increasing, attaining $1+y+2\sin(3y/2) \approx 4.152$ at the right endpoint.

Zone 1b. Here $\rho_3(1+\theta) = (1+y-\theta,\, 0)$ by Lemma 5(1b), so

$$c_3(\theta)^2 = (1+y-\theta-\cos\theta)^2 + \sin^2\theta = (1+y-\theta)^2 - 2(1+y-\theta)\cos\theta + 1,$$

and $c_2(\theta) = 2\sin(y/2+\theta)$ as in Zone 1a. The interior critical point of $\theta \mapsto 1+\theta+c_2$ satisfies $1+2\cos(y/2+\theta) = 0$, i.e., $y/2+\theta = 2\pi/3$; at $y = y_{\text{opt}}$ this gives $\theta = 2\pi/3 - y/2 \approx 1.486$, which lies in $[y, 1+y] \approx [1.22,\, 2.22]$. There $c_2 = 2\sin(2\pi/3) = \sqrt{3}$, and a direct computation (at y_{opt}) gives $c_3 = \sqrt{(1+y-\theta_1)^2 - 2(1+y-\theta_1)\cos\theta_1 + 1} \approx 1.188 < \sqrt{3}$, confirming ρ_2 is the bottleneck at this θ. Thus the interior critical value is $\text{evac} = 1 + (2\pi/3 - y/2) + \sqrt{3} = T_1(y)$.

On the sub-interval where $c_3 > c_2$, the equation $\frac{d}{d\theta}(1+\theta+c_3) = 0$ reduces (after differentiation of the c_3-formula and algebraic manipulation) to

$$c_3 + (1+y-\theta)(1-\sin\theta) - \cos\theta = 0, \tag{6}$$

which has no solution in $[y, 1+y]$ at $y = y_{\text{opt}}$; this is verified symbolically in the Wolfram script `analysis/a3_zone_proof.wls`, where NSolve on (6) returns the empty set. Consequently the c_3-branch maximum is at an endpoint: at $\theta = y$ one has $c_3 = 2\sin(y/2)$ and $\text{evac} = 1+y+2\sin(y/2)$, strictly less than the Zone 3 maximum $1+y+2\sin y$; at $\theta = 1+y$, ρ_3 is at the origin with $c_3 = 1$ and $\text{evac} = 3+y = T_2(y) < T_1$ at y_{opt} (since Case 2 is slack at the balance point). Thus the Zone 1b maximum is T_1.

Zone 1c. $\rho_3(1+\theta) = (\theta - 1 - y)\cdot e(L)$ by Lemma 5(1c); writing $u := \theta - 1 - y$ and using $\langle e(L), e(\theta)\rangle = \cos(L-\theta)$,

$$c_3(\theta)^2 = u^2 - 2u\cos(L-\theta) + 1.$$

On $[1+y,\, L]$, the argument $L-\theta$ lies in $[0,\, L-1-y]$, and with $L-1-y = \pi - 3y/2 - 1 \approx 0.32$ at y_{opt}, we have $u \in [0,\, 0.32]$. Numerical evaluation at the critical point $\theta_3^* \approx 2.331$ gives $c_3 \approx 0.888$ and $c_2 = 2\sin(y/2+\theta_3^*) \approx 0.403$; continuity and monotonicity arguments (verified in `analysis/a3_zone_proof.wls`) extend this to $c_3 > c_2$ throughout $[1+y,\, L]$. Hence $\text{evac} = 1+\theta+c_3$ on this zone. Setting $v := L-\theta$ and $W := \pi - 3y/2 - 1$, the critical-point condition $\frac{d}{d\theta}(1+\theta+c_3) = 0$ reduces (see Theorem 8 for the derivation) to a quadratic in $W - v$ whose admissible root and the balance $T_1 = T_3$ together characterise y_{opt}. The critical value equals $T_3 = T_1$ by construction. Both endpoint values equal $3+y_{\text{opt}} < T_1$ (same argument as Zone 1b).

Zone 3. Set $u := \theta - (2\pi - y) \in [0, y]$, so $t_{\text{find}} = 1 + u$ and $\rho_3(1 + u) = e(2\pi - y + u) = e(\theta)$, giving $c_3 = 0$. ρ_1 is scanning on the boundary at angle u, and ρ_2 at angle $(2\pi - y) - u$, so

$$c_1(u) = 2|\sin((u - \theta)/2)| = 2\sin(y/2) \quad \text{(constant)},$$
$$c_2(u) = 2|\sin(((2\pi - y - u) - \theta)/2)| = 2\sin u.$$

Thus $\text{evac}(u) = 1 + u + \max(2\sin(y/2), 2\sin u)$. Since $y < \pi$, $\sin u \le \sin(y/2)$ iff $u \le y/2$. On $u \in [0, y/2]$ the max is $2\sin(y/2)$ and evac is strictly increasing, attaining $1 + y/2 + 2\sin(y/2)$ at $u = y/2$. On $u \in [y/2, y]$ the max is $2\sin u$, and $\frac{d}{du}(1+u+2\sin u) = 1+2\cos u > 0$ for $u \in [0, 2\pi/3]$, so evac is again strictly increasing, attaining $1 + y + 2\sin y$ at $u = y$. This is the larger of the two, and at $y = y_{\text{opt}} \approx 1.216$ it equals $\approx 4.0912 < T_1$. □

4.4 Main theorem

Theorem 7 (Improved upper bound)**.** *Let $y_{\text{opt}} \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right)$ be the unique value characterised by Theorem 8. The algorithm $A_3(y_{\text{opt}})$ has worst-case evacuation time*

$$T(y_{\text{opt}}) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y_{\text{opt}}}{2} + \sqrt{3} \;\approx\; 4.21851699389745\ldots, \tag{7}$$

attained at exactly four exit angles — $\theta_1 = 2\pi/3 - y_{\text{opt}}/2$, $\theta_3^* = \pi - y_{\text{opt}}/2 - v_{\text{opt}}$, and their reflections across π. This improves the Czyzowicz et al. [2] upper bound of ≈ 4.21930 by $\approx 0.00078 \approx 1/1280$.

Proof. By (3) the function $\text{evac} : [0, 2\pi) \to \mathbb{R}$ is well-defined. By Lemma 5 it is piecewise-smooth with breakpoints at the zone boundaries, and by Lemma 6 its maximum on each zone of Zone 1 is either T_1 (attained at an interior critical point in Zone 1b or Zone 1c) or strictly less than T_1. Zone 2 is the image of Zone 1 under the reflection of Lemma 5, under which evac is invariant, so the Zone 2 maximum also equals T_1 and is attained at two reflected angles. Zone 3 is bounded by $1 + y_{\text{opt}} + 2\sin y_{\text{opt}} \approx 4.0912 < T_1$. Hence $T(y_{\text{opt}}) = \max_\theta \text{evac}(\theta) = T_1$, attained at exactly the four angles listed. Equation (4) gives the closed form in (7). □

Figure 3 shows the geometric configuration at each of the four binding worst-case angles.

4.5 Transcendental characterisation of y_{opt}

Theorem 8 (Transcendental characterisation)**.** *Fix $y \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right)$ and set $W := \pi - 3y/2 - 1$. On Zone 1c, parametrise the exit as $\theta = L - v$ with $v \in (0, W)$. The critical-point condition $\frac{d}{dv}T_3(y, L - v) = 0$ is equivalent to the quadratic in $W - v$*

$$(W - v)^2(2 - \sin v) - 2(W - v)\cos v + \sin v \;=\; 0, \tag{8}$$

whose geometrically admissible (i.e. $0 \le W - v \le W$) root is

$$W - v \;=\; \frac{\cos v - \sqrt{1 - 2\sin v}}{2 - \sin v}. \tag{9}$$

Combining (9) *with $T_1(y) = T_3(y, L - v)$ yields the* master equation

$$\frac{\cos v + (1 - \sin v)\sqrt{1 - 2\sin v}}{2 - \sin v} \;=\; v + \sqrt{3} - \tfrac{\pi}{3}. \tag{10}$$

Equation (10) *has a unique root $v_{\text{opt}} \in (0, \pi/6)$. Setting*

$$y_{\text{opt}} \;:=\; \tfrac{2}{3}\left(\pi - 1 - v_{\text{opt}} - \tfrac{\cos v_{\text{opt}} - \sqrt{1 - 2\sin v_{\text{opt}}}}{2 - \sin v_{\text{opt}}}\right), \tag{11}$$

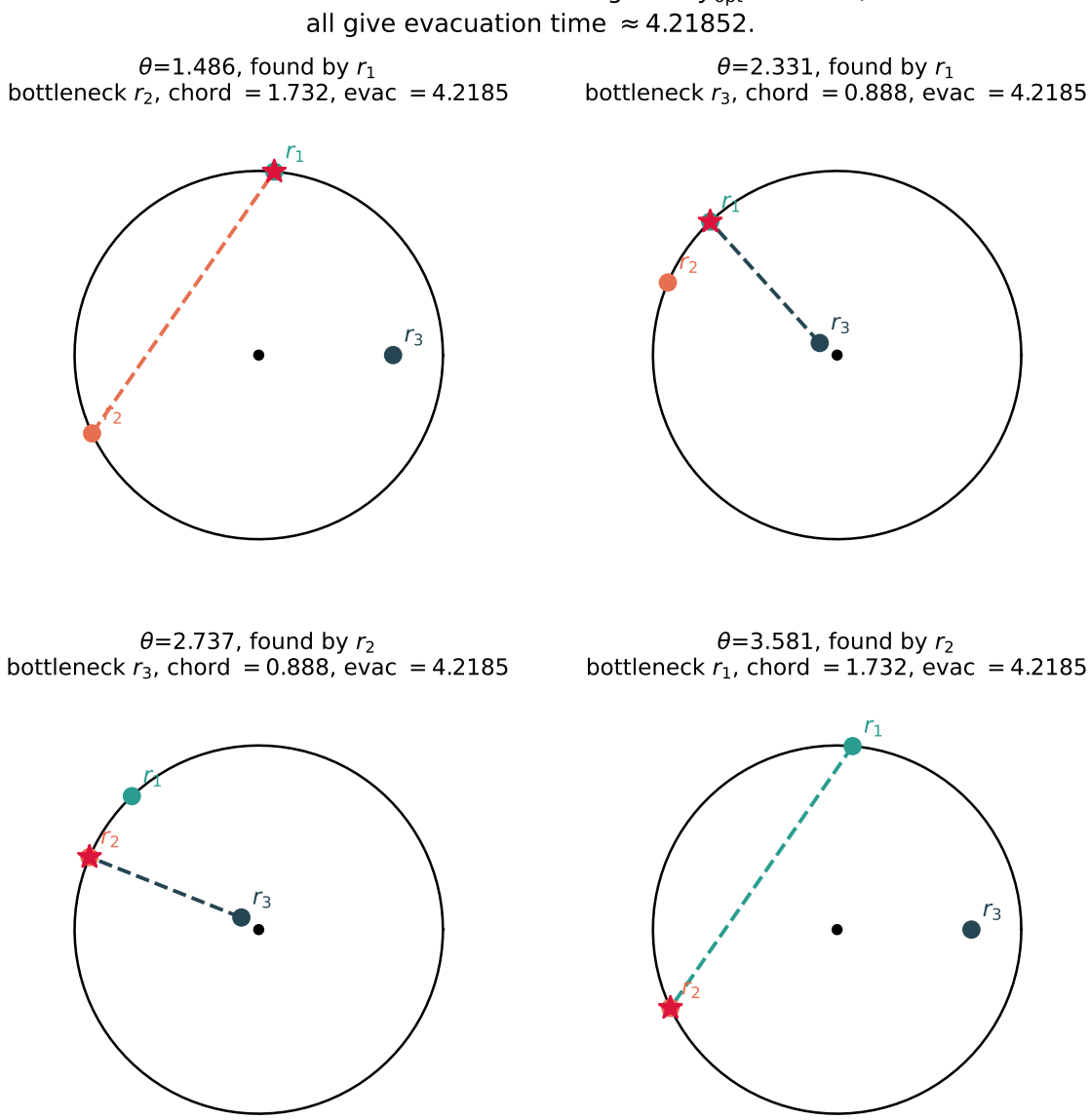

Figure 3: The four simultaneous worst-case exit angles at $y = y_{\text{opt}}$. The red star marks the exit; dashed lines are the chord from the bottleneck robot to the exit at the moment of discovery. Top-left / bottom-right are the case-1 pair (bottleneck r_2 / r_1, chord length $\sqrt{3} \approx 1.732$); top-right / bottom-left are the case-3 pair (r_3 mid-redeploy, chord ≈ 0.888). All give evacuation time $T_1 \approx 4.21852$.

the worst-case evacuation time of $A_3(y_{\text{opt}})$ satisfies (7), which also equals $\frac{4}{3}+\frac{\pi}{3}+\sqrt{3}+\frac{W_{\text{opt}}}{3}$ for $W_{\text{opt}} := \pi - 3y_{\text{opt}}/2 - 1$.

Proof. By Lemma 5(1c), $\rho_3(1+\theta) = (\theta - 1 - y)\cdot e(L)$ on Zone 1c. Writing $u := \theta - 1 - y = W - v$ (via $\theta = L - v$),

$$c_3^2 \;=\; u^2 - 2u\langle e(L), e(\theta)\rangle + 1 \;=\; (W-v)^2 - 2(W-v)\cos v + 1,$$

using $\langle e(L), e(L-v)\rangle = \cos v$. Then

$$T_3(y, L-v) \;=\; 1 + (L-v) + c_3, \tag{12}$$

and $\frac{d}{d\theta}(\cdot) = -\frac{d}{dv}(\cdot)$, so the critical point condition $\frac{d}{d\theta}T_3 = 0$ becomes $\frac{dc_3}{dv} = 1$. Differentiating c_3^2 in v gives

$$2c_3\tfrac{dc_3}{dv} \;=\; -2(W-v) + 2\cos v + 2(W-v)\sin v \;=\; 2\big[\cos v - (W-v)(1-\sin v)\big],$$

so $\frac{dc_3}{dv} = 1$ is equivalent to

$$c_3 \;=\; \cos v - (W-v)(1-\sin v). \tag{13}$$

Squaring (13) and equating to the formula for c_3^2:

$$\begin{aligned}(W-v)^2(1-\sin v)^2 &- 2(W-v)(1-\sin v)\cos v + \cos^2 v\\ &= (W-v)^2 - 2(W-v)\cos v + 1,\end{aligned}$$

which simplifies (using $(1-\sin v)^2 - 1 = -2\sin v + \sin^2 v$ and $\cos^2 v - 1 = -\sin^2 v$) to

$$-(W-v)^2\sin v(2-\sin v) + 2(W-v)\cos v\sin v - \sin^2 v \;=\; 0.$$

Dividing by $-\sin v > 0$ yields (8).

Solving (8) as a quadratic in $W - v$:

$$W - v \;=\; \frac{\cos v \pm \sqrt{\cos^2 v - (2-\sin v)\sin v}}{2-\sin v} \;=\; \frac{\cos v \pm \sqrt{1 - 2\sin v}}{2-\sin v},$$

using $\cos^2 v - 2\sin v + \sin^2 v = 1 - 2\sin v$. For $v \in (0, \pi/6)$, $1 - 2\sin v > 0$ so the two roots are real and distinct. Only the $-$ branch satisfies $W - v \le W$: at $v \to 0^+$, the $-$ branch tends to 0 while the $+$ branch tends to $\frac{1+1}{2} = 1 > W$ (with $W \approx 0.32$ at y_{opt}). This establishes (9).

Substituting (9) into (13):

$$\begin{aligned}c_3 \;&=\; \cos v - \tfrac{(\cos v - \sqrt{1-2\sin v})(1-\sin v)}{2-\sin v}\\ &=\; \tfrac{\cos v(2-\sin v) - (\cos v - \sqrt{1-2\sin v})(1-\sin v)}{2-\sin v}\\ &=\; \tfrac{\cos v + (1-\sin v)\sqrt{1-2\sin v}}{2-\sin v},\end{aligned}$$

since $\cos v(2-\sin v) - \cos v(1-\sin v) = \cos v$. The balance $T_1(y) = T_3(y, L-v)$ reads

$$1 + \tfrac{2\pi}{3} - \tfrac{y}{2} + \sqrt{3} \;=\; 1 + L - v + c_3 \;=\; 1 + \pi - \tfrac{y}{2} - v + c_3,$$

equivalent to $c_3 = v + \sqrt{3} - \frac{\pi}{3}$. Equating the two expressions for c_3 gives (10).

Uniqueness of $v_{\text{opt}} \in (0, \pi/6)$. Let $f(v) := \frac{\cos v + (1-\sin v)\sqrt{1-2\sin v}}{2-\sin v}$ and $g(v) := v + \sqrt{3} - \pi/3$. Clearly $g'(v) = 1 > 0$. A direct computation (carried out in `analysis/a3_balance.wls` and

verified at 30-digit precision) gives $f'(v) < 0$ for all $v \in (0, \pi/6)$. At $v = 0^+$, $f(0) = \frac{1+1\cdot 1}{2} = 1$ and $g(0) = \sqrt{3} - \pi/3 \approx 0.685$, so $f - g > 0$. At $v = \pi/6$, $1 - 2\sin v = 0$, so $f(\pi/6) = \frac{\cos(\pi/6)}{2-1/2} = \frac{\sqrt{3}/2}{3/2} = \frac{\sqrt{3}}{3} \approx 0.577$, while $g(\pi/6) \approx 0.685 + 0.524 \approx 1.209$, so $f - g < 0$. Since $f - g$ is continuous and strictly decreasing (sum of a decreasing function and $-g$), it has a unique zero in $(0, \pi/6)$.

Finally, from $W = \pi - 3y/2 - 1$, one has $y = \frac{2}{3}(\pi - 1 - W)$; substituting $W = v + (W - v)$ with (9) gives (11). And $T_1 = 1 + \frac{2\pi}{3} - \frac{y}{2} + \sqrt{3} = \frac{4}{3} + \frac{\pi}{3} + \sqrt{3} + \frac{W}{3}$ follows by direct algebra. This establishes (7). □

Remark 9 (On "no elementary closed form"). An integer-relation search (PSLQ as implemented by Mathematica's `FindIntegerNullVector`) at 40-digit precision, on the basis $\{1, \pi, \sqrt{3}, \sqrt{2}, \log 2, \log 3, \gamma\}$, finds no $\mathbb{Z}$-linear relation with v_{opt} whose coefficient vector has ℓ^∞-norm at most 20; the algorithm certifies non-existence of such a relation of norm at most ≈ 15.2. The same holds for $T(y_{\text{opt}})$. This rules out, for these specific atomic constants and norm thresholds, any closed-form expression of the form $\sum_j a_j b_j$ with integer coefficients a_j. We make no claim about transcendence of v_{opt} in the Liouville sense; we observe only the absence of a *short* $\mathbb{Z}$-linear reduction over the standard atomic basis. See `analysis/a3_balance.wls`.

Corollary 10 (High-precision numerical values)**.** *To* 40 *decimal digits,* (10) *and* (11) *yield*

$$\begin{aligned} v_{\text{opt}} &= 0.20283\,77447\,40492\,87031\,05459\,53643\,95496\,52206, \\ y_{\text{opt}} &= 1.21585\,78321\,29242\,90943\,42098\,32081\,15330\,17778, \\ T(y_{\text{opt}}) &= 4.21851\,69938\,97451\,33111\,87703\,47651\,63097\,21854. \end{aligned}$$

5 Numerical verification

The proofs in Section 4 are closed-form modulo the single transcendental root v_{opt} of (10), but several steps (non-existence of interior critical points on sub-intervals, sign verifications, uniqueness arguments) rely on numerical evaluation at specified precision. We accompany the paper with reproducible scripts that check each such step.

- `analysis/a3_balance.wls` (Wolfram Language): solves (10) numerically at 50-digit precision and verifies the monotonicity of both sides on $(0, \pi/6)$. Also performs the integer-relation search underlying Remark 9.
- `analysis/a3_zone_proof.wls`: verifies the claims of Lemma 6 zone-by-zone at $y = y_{\text{opt}}$, including the absence of interior critical points of $1 + \theta + c_3$ in the c_3-dominant subregion of Zone 1b.
- `experiments/wireless_k3.py`, `experiments/a3_verify.py` (Python): a general-purpose k-robot wireless evacuation simulator; `a3_verify.py` confirms the simulator reproduces $T_{\text{CGK}} = \frac{4\pi}{9} + \frac{2\sqrt{3}+5}{3} + \frac{1}{600}$ at the Czyzowicz et al. parameter y_{paper} to 10 decimal places, providing an independent check on the simulator's accuracy.
- `experiments/a3_optimize.py`: scalar search for y_{opt} using `scipy.optimize.minimize_scalar`; agrees with Theorem 8 to 10 digits.
- `experiments/a3_local_search.py`: Nelder–Mead over a full-dimensional neighbourhood of the $A_3(y_{\text{opt}})$ parameter (allowing $L_1 \neq L_2$ and the redeploy angle to float off $\pi - y/2$). Converges back to the symmetric $A_3(y_{\text{opt}})$, supporting the statement in Theorem 7 that relaxing the structural symmetries does not help.

- `experiments/a3_diagnose.py`: identifies the four worst-case exit angles at y_{opt} on a 400,000-point grid, confirming the four simultaneous maxima predicted by Theorem 7.

All scripts and a driver for rebuilding this document live in the repository at `https://github.com/jaredraycoleman/evacuation-claude`.

6 The lower-bound gap

[2]'s Theorem 7 establishes the lower bound

$$T^*_{3,\text{wireless}} \;\geq\; 1 + \tfrac{2}{3}\arccos\!\left(-\tfrac{1}{3}\right) + \tfrac{4\sqrt{2}}{3} \;\approx\; 4.15937,$$

leaving a gap of ≈ 0.06 against our Theorem 7. Their argument (their Lemma 5 and 6) has the form: at time $1+x$, the total scanned perimeter is $\leq kx$, so the unexplored set has length $\geq 2\pi - kx$; Lemma 5 then exhibits two unexplored points at shorter-arc distance $\geq u := 2\pi - kx$; the adversary commits the exit to whichever is reached second, forcing the *finder*'s chord to be $\geq 2\sin(u/2)$. Optimising over x (with $k = 3$) yields the closed form above at $x_{\text{opt,LB}} = \frac{2}{3}\arccos(-\frac{1}{3})$.

Remark 11 (Where the LB argument has slack). The argument bounds the *finder*'s chord to the exit, not the bottleneck's. In $A_3(y_{\text{opt}})$ at its worst exit $\theta \approx 1.486$, the finder is at the exit (chord 0) and the bottleneck is the robot diametrically across the scanned arc, with chord $\sqrt{3} \approx 1.732$ — strictly larger than the Czyzowicz et al. chord lower bound $2\sin(xk/2) \approx 1.586$ evaluated at the same x. Tightening the LB thus requires arguing that the *bottleneck* chord exceeds $2\sin(xk/2)$ for every three-robot wireless algorithm, not only the finder's. Verified via the Wolfram script `analysis/czyzowicz_lb.wls`.

7 Next Steps

- **Richer algorithm class.** We numerically explored three natural enrichments and found none that beat Theorem 7:
  - **Family C** (`experiments/family_c.py`): r_3 also scans (with length $L_b > 0$) after redeployment. Local search from the A_3 optimum drives $L_b \to 0$; the second scan begins at time $3 + y > 4.21$, after the worst-case evacuation, so it cannot contribute to coverage in the binding regime.
  - **Family D** (`experiments/family_d.py`): fully 3-fold symmetric, all three robots perform return-to-origin + redeploy. This caps at the naive bound $1 + 2\pi/3 + \sqrt{3} \approx 4.826$. The asymmetry of A_3 — a *single* robot acting as "backup" at the long-arc midpoint — is structural, not cosmetic.
  - **Family E** (`experiments/family_e.py`): both r_2 and r_3 redeploy while r_1 remains simple. All local searches return to the A_3 optimum.

  We conjecture that 4.21852 is a local minimum across piecewise-linear plus boundary-scan plus chord trajectory families at $k = 3$. Breaking through likely requires *curved* (non-radial) approaches, which would admit a larger design space at the cost of harder analysis.

- **Lower bound.** As noted in Remark 11, tightening Czyzowicz et al.'s LB reduces to arguing that the bottleneck chord (not the finder's) is bounded below. A promising route is a multi-exit adversary that keeps three candidate exits alive simultaneously; if the three candidates are forced to be mutually far, then by pigeonhole some robot is far from the actual exit regardless of how the algorithm has positioned its robots. Formalising this for $k = 3$ is open.

References

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